

Exercise 6: Library Management System

Scenario

You are developing a library management system where users can search for books by title or author.

1. Understand Search Algorithms

Linear Search

Linear Search is a simple searching technique in which each element is checked sequentially until the required element is found or the end of the collection is reached.

Characteristics

- Works on both sorted and unsorted data.
- Easy to implement.
- Suitable for small datasets.

Binary Search

Binary Search is an efficient searching algorithm that repeatedly divides a sorted collection into two halves until the required element is found.

Characteristics

- Requires the data to be sorted.
- Faster than Linear Search for large datasets.
- Uses the divide-and-conquer approach.

2. Setup

A Java project named `LibraryManagementSystem` was created.

The project contains a code folder with the following files:

`LibraryManagementSystem`

└── `code`

 └── `Book.java`

 └── `LibraryManagementSystem.java`

- `Book.java` – Defines the `Book` class with attributes `bookId`, `title`, and `author`.
- `LibraryManagementSystem.java` – Implements Linear Search and Binary Search to search books by title.

3. Implementation

Book Class

The Book class contains the following attributes:

- bookId
- title
- author

Search Operations

The following searching algorithms are implemented:

- Linear Search by title
- Binary Search by title (after sorting the array alphabetically)

4. Analysis

Time Complexity Comparison

Search Algorithm	Best Case	Average Case	Worst Case
Linear Search	$O(1)$	$O(n)$	$O(n)$
Binary Search	$O(1)$	$O(\log n)$	$O(\log n)$

When to Use Each Algorithm

Linear Search

- Suitable for small datasets.
- Can search both sorted and unsorted data.
- Easy to implement.

Binary Search

- Suitable for large datasets.
- Requires the data to be sorted before searching.
- Much faster than Linear Search for repeated searches.

5. Conclusion

The Library Management System was successfully implemented using both Linear Search and Binary Search algorithms. Linear Search provides flexibility by working with unsorted

data, whereas Binary Search offers significantly better performance for large sorted datasets. Therefore, Binary Search is the preferred choice when the collection is maintained in sorted order.